

AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

FLIGHT PERFORMANCE ANALYSIS OF THE NOVEL SYNCHROCOPTER



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Abstract

This third year Group Design Project (GDP) aims to develop a preliminary design for the mid-air recovery aspect of a wholly reusable launch system for which as much as possible can be recovered and reused for subsequent launches of LEO satellites. This report forms a subset of the marking criteria for the *AERO 96005 - Group Design Project* module, seeking to test engineering judgement and design tradeoffs within group settings.

This report outlines my main contributions to the project as an engineer from the Flight Mechanics, Performance and Controls team, as well as my notable contributions to other areas within the project. My main contribution to the project is analysing the performance of the helicopter and ensuring that it adheres to the initially-laid standards. I have analysed the rate of climb, best climb speed, rate of descend, maximum forward flight speed with the slung payload, the power vs airspeed, the best range and endurance speeds, the fuel-to-climb, range and endurance, operational radius and the hover ceiling. These metrics show that the helicopter is capable of fulfilling the initially-laid standards and completing the mission. The results are summarised in the table below. As for my other contributions, I helped formulate the free-body diagram and the blade flapping angle part of the state space mode. I also consulted for the empennage sizing and fuselage design, and performed a trade-off study between LQG and MPC Control for our helicopter.

Performance Metric	Value	Remarks
Max Rate of Climb at GTOW	37.74 m s ⁻¹	
Best Climb Speed at GTOW	20.12 m s ⁻¹	
Max Rate of Climb at MTOW	16.89 m s ⁻¹	
Best Climb Speed at MTOW	32.31 m s ⁻¹	
Rate of Descend	9.4 m s ⁻¹	Achievable without using autorotation and without onset of VRS.
Max Forward Flight Speed with slung payload	30 m s ⁻¹	
Best Range Speed GTOW @20,000 ft	62.48 m s ⁻¹	
Best Endurance Speed GTOW @20,000 ft	26.21 m s ⁻¹	
Best Range Speed MTOW @3,000 ft	30 m s ⁻¹	Limited by max forward flight speed with payload.
Best Endurance Speed MTOW @3,000 ft	30 m s ⁻¹	Same as above.
Fuel-to-climb	13.25 kg	
Fuel required by mission	202.55 kg	Initial estimation is 273.7 kg.
Maximum range boost	54.3 km	Using all extra fuel in 2nd cruise.
Maximum endurance boost	1810 s	Using all extra fuel in last hover.
Hover ceiling with mission payload @ISA temp.	14,000 ft	
Hover ceiling with mission payload @ISA+20K temp.	12,000 ft	
Maximum payload @20,000 ft @ISA temp.	942 kg	
Maximum payload @10,000 ft @ISA temp.	1753 kg	

WRR Team Members

Table 1: *Team Members and Roles.*

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra Atri Sharma Alastair Wood	Structures & CAD	WRR03
Omar Arangath Thomas Cross Zuzanna Janusz Sam Jenner Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai San Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
Tom Alston Hakan Serpen Tze Leung Wong	Flight Simulation & Testing	WRR06

†Sub-team leads are shown in **bold**

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Part I

Introduction

Wholly or partially reusable launch systems have captured the minds of many when envisioning a cost sustainable solution for multi-mission space transportation. Notable attempts include the NASA 'Space Shuttles', the Soviet Buran programme, and more recently, the SpaceX Falcon 9 rockets.

This project aims to develop an aircraft to capture the launch system while descending back to Earth for reuse. The mode of recovery and transport chosen for this mission is a remote-controlled twin-intermeshing-rotor helicopter, similar to that of the Kaman K-Max.

This report details my main contributions to the project as an engineer from the WRR05 team, and outlines my notable contributions to other areas within the project.

Part II

Pre-Conceptual: Twin Tiltrotor with Skycrane-like Fuselage Concept Proposal

It is deemed early on in the project that vertical take-off capabilities from a vessel on sea is crucial for the recovery of our rocket as missions like these typically occur over sea, not land. I was part of the group to perform a conceptual design for a tiltrotor aircraft, and conduct a trade-off study for the use of tiltrotors against a helicopter or a fixed-wing aircraft for our mission.

I proposed that we create a twin-engine tiltrotor with a Skycrane-like fuselage design concept. The aircraft would have the tiltable engines situated at the wingtips of a low aspect ratio wing. This has effect of balancing the torque exerted on the wing and the aircraft, as well as cancelling out the tip vortices of the wing, allowing for a much lower aspect ratio wing than if the engines are placed mid-wing as generally a low aspect wing would have large induced drags generated by the tip vortices, and would hence be very inefficient in cruise, if the vortices are left unchecked.

The other main feature of this craft is its Skycrane-like fuselage. This minimises the weight of the fuselage, leading to more excess power and hence higher lifting capabilities. A winch would then be used to reel the payload towards the aircraft to minimise drag in forward flight and cruise back to base. This type of fuselage also has the added benefit of the ability to carry awkwardly shaped cargo due to its lack thereof a fuselage, which is a highly probable situation considering that our mission requires the aircraft to bring back rocket parts of different sizes back to base.

The aircraft would have the following specifications.

Empty weight (kg)	2900
Maximum Speed	200 kmph (hover); 500 kmph (forward flight)
Service ceiling	20000 ft (hover); 30000 ft (forward flight)
Range	1500 km
Power required at 20,000 ft (kW)	4400
Diameter of each rotor (m)	5.6
Power to weight ratio	6.67

The tiltrotor concept has been dropped starting from Week 2 in favour of a helicopter due to its lack of apparent advantages over a helicopter in the prescribed mission profile.

Part III

Preliminary Design

1 Main Contribution

My main contribution to this project is taking charge of analysing the performance of the helicopter, and verifying that it meets the prescribed performance standards during the post-design stage.

It should be noted that the performance analyses assume that our synchrocopter behaves similarly to a conventional helicopter, and hence the performance analysis methods of conventional helicopters are used, as from a performance aspect, the two are not too different in principle.

1.1 Field Performance: H-V Diagram with Take-Off and Landing Profile

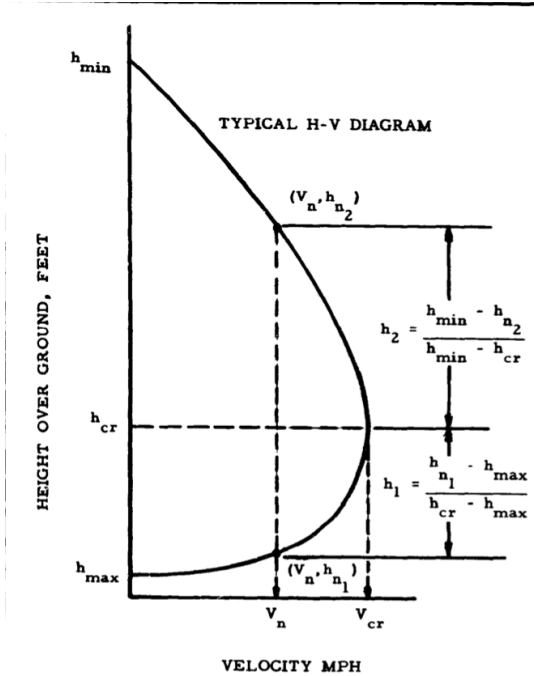


Figure 1: A typical HV Diagram of a helicopter with typical nomenclature [1].

The field performance analysis conducted below gives the combinations of altitude and forward flight speed that a pilot of the synchrocopter will need to avoid due to the onset of the Vortex Ring State (VRS), and the combinations the pilot would be recommended to follow during take-off and landing. The calculation of size of the landing pad on the vessel required is not included, as it is limited by regulation and directly affects the size of the vessel, and hence the mission cost. As such, this responsibility is delegated to the project coordinators [4].

Although both analytical approaches [5] and empirical methods [1] are considered when formulating our Height-Velocity (H-V) diagram, only the empirical method has been used to formulate the H-V Diagram due to the analytical method being overly complex for our purposes.

As seen in Figure 1, the points of interest in the H-V diagram are h_{max} , h_{min} , h_{cr} and v_{cr} . As mentioned and analysed in [1], h_{max} and h_{cr} are reasonably consistent across all helicopters, at around 10 feet and 100 feet respectively. Due to the lack of relevant test data for our synchrocopter at this design stage, those numbers for h_{max} and h_{cr} are assumed. Furthermore, h_{cr} is empirically related

to v_{cr}^2 by the equation [1]

$$h_{min} = 200 + 0.1336 \cdot v_{cr}^2,$$

therefore, all we need is a data point for v_{cr} , which is found to be around 10 knots (11.85 pmh) through flight tests by the simulations team [6] (Note that the units of v_{cr} in this empirical relation is mph). The curve is then drawn by approximating the lower half of the curve to be an ellipse, and the upper part of the curve to be a parabola. The results are shown in Figure 2.

The recommended take-off and landing profile is then drawn after consulting Figure 3 from [2]. The H-V diagram shows that sample take-off/landing profile is around 10 knots (around 10 mph) to the right of

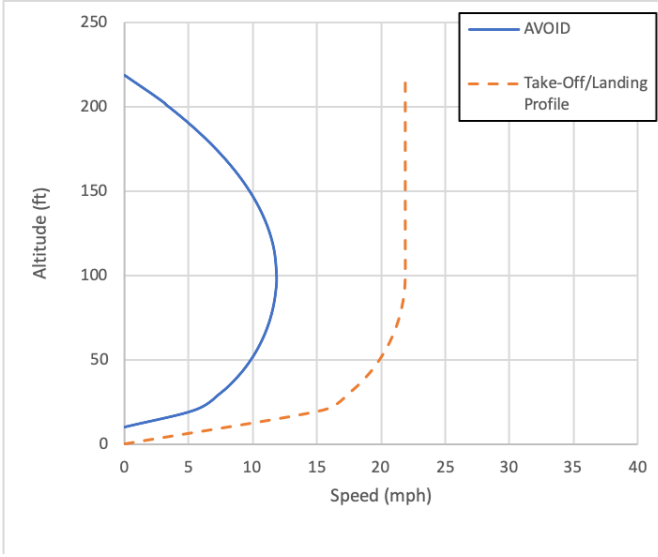


Figure 2: H-V Diagram of our synchrocopter.

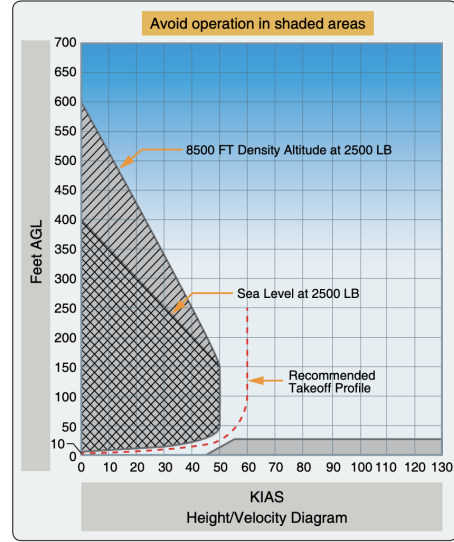


Figure 3: H-V diagram for a Robinson Model R44 II [2].

the tip of the 'AVOID' curve, hence the same is done for our H-V Diagram as a rough estimate of the take-off/landing profile for our synchrocopter.

It should be noted that the H-V diagram produced using this method should only serve as an initial estimate. For additional accuracy, either an analytical method or flight testing for the entire profile should be carried out.

1.2 Point Performance

The point performance section includes the Rate of Climb vs Speed graph during take-off for our synchrocopter as it is a typical metric to use to judge the point performance. The Descend Rate vs Speed graph is also included as our mission profile requires a high descend rate manoeuvre to capture the rocket, hence making this a pertinent metric to analyse.

1.2.1 Rate of Climb vs Speed

The rate of climb (ROC) vs speed graphs shows the speed at which the maximum rate of climb is attainable, which is then used to climb to the cruising altitude. This is important since our mission is a time sensitive mission, hence getting to the cruising altitude as soon as possible would be in our best interests.

ROC depends on the excess power at the given speed and altitude, which is the power required for level flight P_{level} subtracted from the maximum power available P_e . The rate of climb is then calculated by [7]

$$\frac{dh}{dt} = \frac{P_e(U, h) - P_{level}(U, h)}{W} - \frac{D}{W}$$

where W is the weight of the helicopter. Furthermore, the power losses due to onset of drag as the forward velocity of the helicopter increases needs to be accounted for as well using the drag coefficient of 0.418 m^2 provided by the aerodynamics team [8].

The analysis is carried out at sea level conditions and the results are shown in Figure 4. It should be noted that the vertical climb rates are generally not the most efficient manner to climb for a helicopter. For vertical climb rate results please refer to [9].

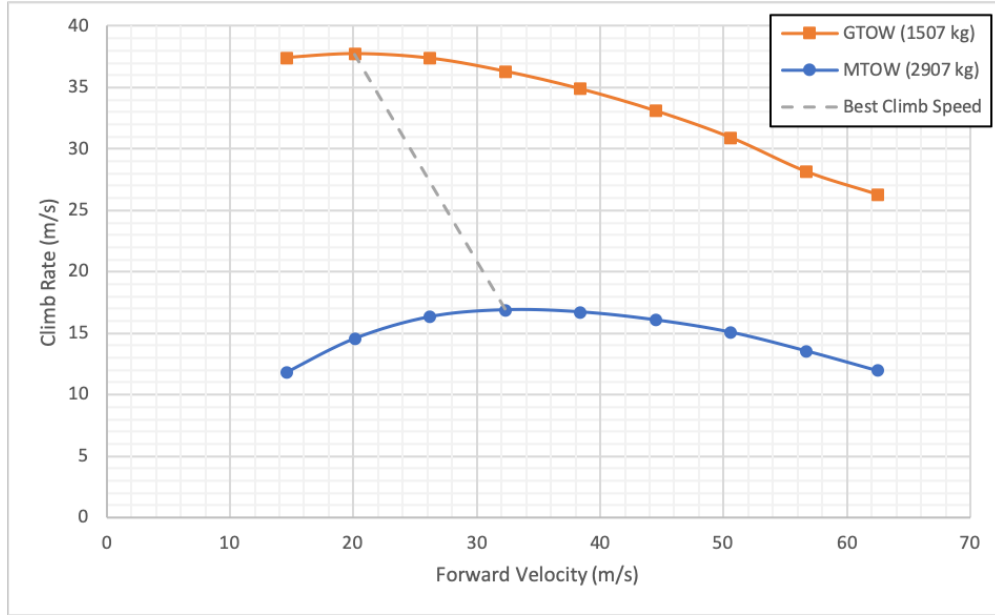


Figure 4: Rate of Climb vs Speed at GTOW and MTOW.

The results of ROC calculated using GTOW showed some unreasonably high values a maximum climb rate of nearly 40 m s^{-1} , while the maximum climb rate averaged by the world record of fastest climb by a helicopter averaged around 17 m s^{-1} [10]. An investigation has been carried out and two main reasons as to why our helicopter has such a high climb rate has been identified:

1. The assumptions regarding the power generated in the overlapping areas of the intermeshing rotors are different between the aerodynamics team and the propulsion team: The propulsion team assumed and sized the power of the engine by assuming a detriment of power in the overlapping areas, while the aerodynamics team has assumed the rotors to behave like they would in tandem (i.e. no power loss in the overlapping areas). After this is discovered, the aerodynamics team and propulsion team unified the power detriment in the overlapping areas using the correction factor in Leishmann [11], the results from the two teams have agreed.
2. The propulsion team has sized the power of the helicopter to take off at MTOW (2907 kg), which is not the case for our mission, as we take off with GTOW (1507 kg) and then capture the payload mid-flight. This is why our helicopter has so much excess power at sea level as whilst not carrying any payload, the mass of the helicopter has been almost halved during take off. Hence, the results for taking off at MTOW are computed and shown in Figure 4, which shows comparatively reasonable results for climb rates.

1.2.2 Rate of Descend vs Speed

The purpose of analysing rate of descend (ROD) is to show that our helicopter would not be caught in the VRS. While it is relatively straightforward to analyse rate of climb, it is excruciatingly difficult to analyse the ROD, especially the size of the VRS envelope. Hence, only empirical methods are employed in analysing whether our helicopter would be caught in VRS.

There have been many attempts in calculating the VRS envelope, the most notable of which is the Meijer Drees model, which is used in this analysis. Other models include the Washizu, Peters and Leishman, all the simulation data of which are included in [12]. There is also a overly conservative 'rough' profile provided in [3], which is a half-circle that nearly encompasses all of the above mentioned profiles, and this will also be used in the analysis for illustrative purposes.

The graph of ROD vs Speed is first non-dimensionalised by induced velocity at hover, $v_i = 20.7035 \text{ m s}^{-1}$, the details of which could be found in [9]. Then the rough and Meijer Drees VRS, and the autorotation curve from [3] in Appendix C are empirically sized in the diagram. The descend profile chosen, which is a 15 descend, is then plotted onto the same diagram as well to see if the descend profile would coincide with the VRS envelope. The results are shown in figure 5.

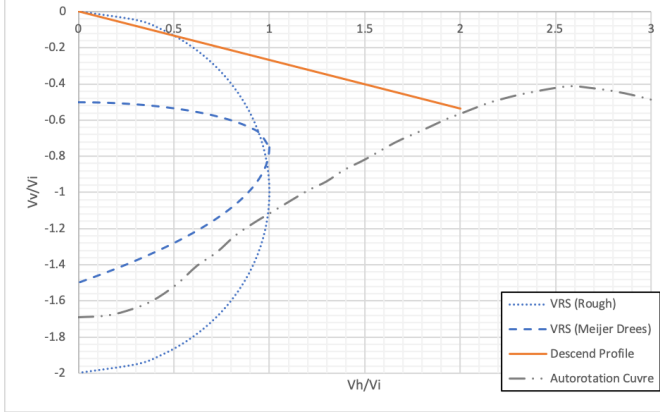


Figure 5: Non-dimensionalised ROD vs Speed graph.

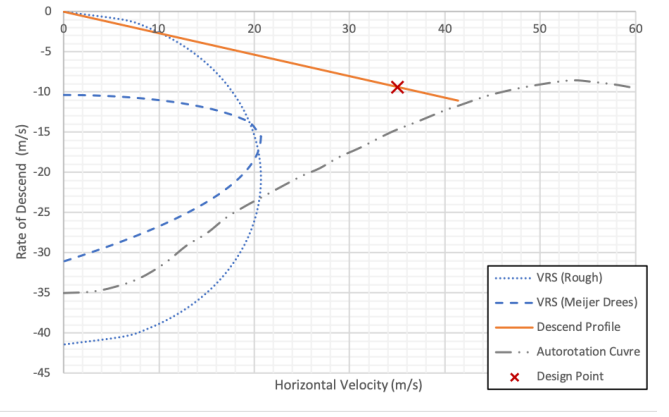


Figure 6: ROD vs Speed graph.

The diagram is then dimensionalised to show the true sizes of the VRS envelopes, and to illustrate where our design point of horizontal velocity of 35 m s^{-1} lies in the diagram in figure 6.

The results show that the descend profile does not coincide with the Meijer Drees VRS envelope. The descend profile has also been verified to not coincide with the other profiles in [12] by the means of observation using a ruler. The descend profile overlaps slightly with the rough VRS envelope during the start of the descent, but this should not be a problem as the rough VRS is an overly conservative estimate.

It should also be noted that our design point is above the autorotation curve in figure 6, which is the various combinations of horizontal and vertical speeds that supply the required energy to keep the rotor turning [2] during an unpowered descend. Hence, there is no need for the use of autorotation to conduct the descent manoeuvre while capturing the payload.

1.3 Mission Performance

1.3.1 Maximum Forward Flight Speed with Slung Payload

The limiting factor for the maximum forward speed with the presence of the slung payload is conjectured to be the instability of the helicopter-payload system as a whole, instead of the onset of supersonic effects at the advancing side of the rotor blade, which is conjectured to be the case for the helicopter at its maximum forward flight speed without the slung payload, details of which are in [13].

However, the initial conjecture has been proven wrong using an eigenvalue analysis of the A matrix conducted by the Flight Mechanics team [14], and from Padfield [15], who state that helicopter modes behave more like fixed-wing aircraft modes and the coupling effects decrease in forward flight. The dynamic stability analysis is not complete due to the lack of time, but it points that cruising at 30 m s^{-1} with a slung payload at 3,000 ft should be reasonably stable, and this is confirmed by the results of the simulations team [16]. Hence, 30 m s^{-1} is the cruise speed chosen for the helicopter during 2nd cruise until a more comprehensive set of static and dynamic stability analyses are available.

1.3.2 Power vs Airspeed, and Best Range and Endurance speed

The plot of Power vs Airspeed at different altitudes is essential, as it would provide a mean to obtain the best endurance speed (the speed at minimum power) and the best range speed (the speed at which the curve is tangent to a straight line drawn from origin) [7]. The altitudes of 20,000 ft, 10,000 ft and 3,000 ft are chosen because they are the cruising altitude, mid-climb altitude and the altitude at which the helicopter would cruise for a short time after capturing the payload respectively. Using the rotor code from the aerodynamics team [9], the results are shown in figures 7 and 8.

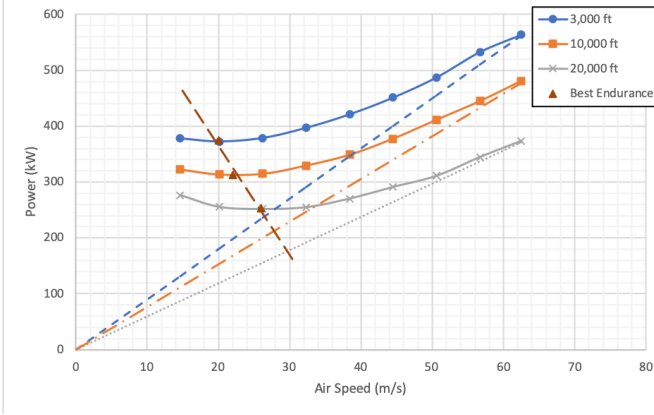


Figure 7: Power vs (True) Airspeed at GTOW.

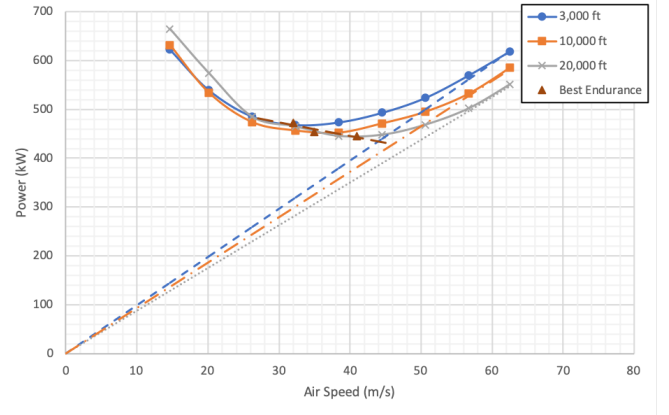


Figure 8: Power vs (True) Airspeed at MTOW.

The results show that as altitude increases, the speed of best endurance increases. Straight trendlines are included in the plot for the purpose of illustrating the trend and for first order approximations only, as it isn't clear if those trends are actually linear not. Upon further research in [7], the trend and the magnitude of its rate of increase is reasonable given that the x-scale in this graph is in m s^{-1} .

As for the speed of best range, drawing a straight line from origin shows that we are not able to reach the point at which the power curve is tangent to the straight line, resulting in the best range speed in terms of power being limited to 62 m s^{-1} in both cases. This is related to the engine power being oversized for our helicopter as mentioned in the previous section, as evident by the curves in the MTOW (Figure 8) are much closer to the point of tangent than the curves in GTOW (Figure 7, although it should be noted that in both cases, the curve of 20,000 ft is much closer to being tangent with the straight line from origin than that of lower altitudes, meaning that the best range speed gets closer to 62 m s^{-1} as the altitude increases, and actually almost coincides with 62 m s^{-1} during MTOW conditions as shown in Figure 8.

1.3.3 Fuel-to-climb (FTC)

The exact FTC can only be calculated if the flight trajectory is specified in advance [7]. This is not the case for this project, hence the first-order approximation from Fillippone [7] is used.

FTC is related to the difference in energy between the helicopter on the ground and in cruise. The difference in energy is given by [7]

$$\Delta E = mgh + \frac{1}{2}mU^2 + \Delta E^{rot}_k \quad (1)$$

where ΔE^{rot}_k is the difference in rotor kinetic energy, m is the Gross Take Off Weight (GTOW) of the helicopter and h is the cruising altitude of the helicopter. Since the RPM of the rotor remains constant during operation, ΔE^{rot}_k results from a difference in forward speed. Hence, ΔE^{rot}_k is given by

$$\Delta E^{rot}_k = \frac{N}{2}MU^2$$

where N is the number of rotors and M is the mass of each individual rotor blade.

Fuel burn is approximated by [7]

$$\Delta E \approx m_f c_p \overline{\eta_1 \eta_2} \quad (2)$$

where m_f is the FTC, c_p is the specific heat of combustion of the fuel, $\overline{\eta_1}$ is the average thermodynamic efficiency of the gas turbine and $\overline{\eta_2}$ is the average propulsion efficiency, which is given by [7]

$$\eta_2 = \frac{W(U \tan \alpha_T + v_C)}{P_e}$$

where α_T is the angle at which the thrust vector will be offset from the hub axis at cruise, v_C is the Rate of Climb and P_e is the power of the engine in level flight.

m_f is then found by combining Equations 1 and 2 into

$$m_f \approx \frac{mgh + \frac{1}{2}mU^2 + \frac{N}{2}MU^2}{c_p \overline{\eta_1 \eta_2}}$$

The values used for this calculation are

m	h	U	N	M	c_p	$\overline{\eta_1}$	$\overline{\eta_2}$	v_C	α_T	P_e
1507 kg	6096 m	51.4 m s ⁻¹	4	99.7 kg	43.5 MJ kg ⁻¹ [7]	0.45 [7]	0.357	8 m s ⁻¹	4°	480 kW

Note that though from previous sections we have found that our helicopter is capable of very high climb rates without the payload, it would simply not make sense in terms of fuel economy to climb at such a high climb rate. Hence, a typical helicopter climb rate of 8 m s⁻¹ is assumed. $\alpha_T = 4^\circ$ is also a typical helicopter rotor tilt angle obtained from Fillippone [7]. c_p and $\overline{\eta_1}$ are obtained from historical data in Fillippone [7] as well. Furthermore, since the helicopter is flying at half of MTOW, P_e is taken to be half of the power of the engine at 10,000 ft to obtain an average value for η_2 .

Hence, m_f is calculated to be 13.25 kg. Note that this value remains a first order approximate due to the lack of information on the exact climb trajectory and engine performance test/simulation data. m_f should be duly updated when the aforementioned values become available.

1.3.4 Range-Endurance Analysis

With a reasonably accurate value for m_f obtained from the previous section, we could analyse the range and endurance performance of our helicopter using the fuel allocated for each flight phase provided by the systems team [17].

The range for a general helicopter could be calculated by [7]

$$R = \int_e^i SAR \, dm \approx SAR \cdot \Delta m = \frac{1}{SFC} \frac{U}{P} \Delta m,$$

while the endurance for a general helicopter is found by [7]

$$E = \int_e^i SE \, dm \approx SE \cdot \Delta m = \frac{1}{SFC} \frac{1}{P} \Delta m.$$

where SAR is Specific Air Range, SE is the Specific Endurance, Δm is the mass of fuel used during the flight segment, U is the forward flight speed and P is the flight power.

The full set of spreadsheet data and results used in this analysis is affixed to Appendix D due to a lack of space. A reduced set of data are shown in Table 2.

Table 2: Range and Endurance analysis data and results with fuel allocated (Reduced).

Segment	Mass of fuel allocated (kg)	Range (m)	Endurance (s)	Target
Climb	13.250	N/A	N/A	N/A
1st Cruise @Best Range Speed	55.250	111567.724	1785.541	75 km
1st Cruise @Best Endurance Speed		69530.428	2652.821	75 km
Hover for 10 mins	40.200	0.000	1184.482	10 mins
Descend and Capture	56.400	60819.671	1678.710	$v_h = -35$
2nd Cruise	86.500	66022.900	2200.763	75 km
Hover for 5 mins	22.100	0.000	442.020	5 mins

It should be noted that where there is a performance distinction between the 1st cruise at best range speed and best endurance speed found from previous sections, there is no such distinction for the 2nd cruise as the speed is limited by the speed at which the helicopter-payload system would become unstable.

From Table 2, the mass of fuel allocated for the main flight segments are 273.7 kg. From the first order estimate of the systems team, most of the segments have been over-allocated fuel, with the exception of the 2nd cruise phase where more range is desired. Furthermore, the requirement for Descend and Capture is the ability to descend at a horizontal velocity at 35 m s^{-1} and a descent angle of 15° , ergo a vertical velocity of around 9.4 m s^{-1} . Translating this into range and endurance terms, by taking into account the altitude over which the helicopter has to descend, it has to traverse either a range of 19.2 km or an endurance of 550 seconds, whichever one is the more limiting. Using this data, a recommended reallocation of the fuel has been computed and is shown in Table 3 (Full set of data is in Appendix D).

Table 3: Range and Endurance analysis data and results with new fuel allocation (Reduced).

Segment	New fuel allocation (kg)	Old (kg)	Range (m)	Endurance (s)
Climb	13.250	13.25	N/A	N/A
1st Cruise @Best Range Speed	37.150	55.25	75019.572	1200.621
Hover for 10 mins	20.380	40.2	0.000	600.491
Descend	18.500	56.4	19949.715	550.641
2nd Cruise	98.270	86.5	75006.594	2500.220
Hover for 5 mins	15.000	22.1	0.000	300.013

Using this new recommended fuel allocation, the amount of fuel needed for the main flight segment has been reduced down to 202.55 kg. This means that the fuel needed for our mission is down by 71.15 kg than our initial estimate. The extra fuel could be removed to save weight and reduce operational costs, or it could be used to extend the range of the helicopter, which will be covered in the next section.

1.3.5 Operational Radius

In this report, the operational radius is defined as the maximum range that the helicopter could cruise at 3,000 ft after retrieving the payload. It is vital to understand the capabilities of the helicopter during this flight stage in case there is an unexpected delay in the flight time or increase in range between the vessel and the recovery area.

From the previous section, it has been shown that there is an extra 71.15 kg of fuel available for our mission if the fuel tanks of the aircraft were filled to their limits. Assuming that all of the extra fuel is devoted to the 2nd cruise flight segment (i.e. fuel allocated is 169.42 kg), the range and the endurance of our helicopter could be boosted up to 129.3 km and 4310 s respectively. Alternatively, if all the extra fuel is devoted to the segment 'Hover for 5 mins' where the payload is being loaded onto the vessel (i.e. 86.15 kg), the endurance in hover could be boosted up to 1723s (29 mins) instead.

1.3.6 Hover ceiling vs Payload

Understanding the hover ceiling vs payload relation for our helicopter is not only important for us to figure out if the helicopter could carry the various payloads at the designated altitudes in the mission profile, but also important for the project coordinators to figure out what types and weights of payload could our helicopter carry at various altitudes should they need to re-purpose the helicopter [4].

The hover ceiling of the helicopter is obtained by simply calculating the maximum power available at different altitudes, and then relating the power output to thrust and performing a force equilibrium analysis. Since our rotor could only spin at a certain RPM, there are two major factors limit the payload capabilities of the rotor:

1. **Limitation by Stall:** One method to increase the thrust generated by rotor is to increase the angle of attack of the rotor blades while maintaining a constant RPM. Hence, the thrust is limited by the stall angle of the rotor blades.
2. **Limitation by Power:** The other limitation is when the angle of attack of the rotor blades are increasing while being less than the stall angle, the power produced by the engine is not enough to counter the induced drag generated by the rotor blades and sustain the constant RPM. Hence, the maximum available power will limit the maximum angle of attack achievable by the rotor blades.

Using the rotor code of the aerodynamics team, it is found that our payload is entirely limited by power (For more details, please refer to [9]). A graph of total helicopter mass vs altitude is obtained [9], and the payload results are subsequently obtained by subtracting the empty weight of the helicopter from total helicopter,

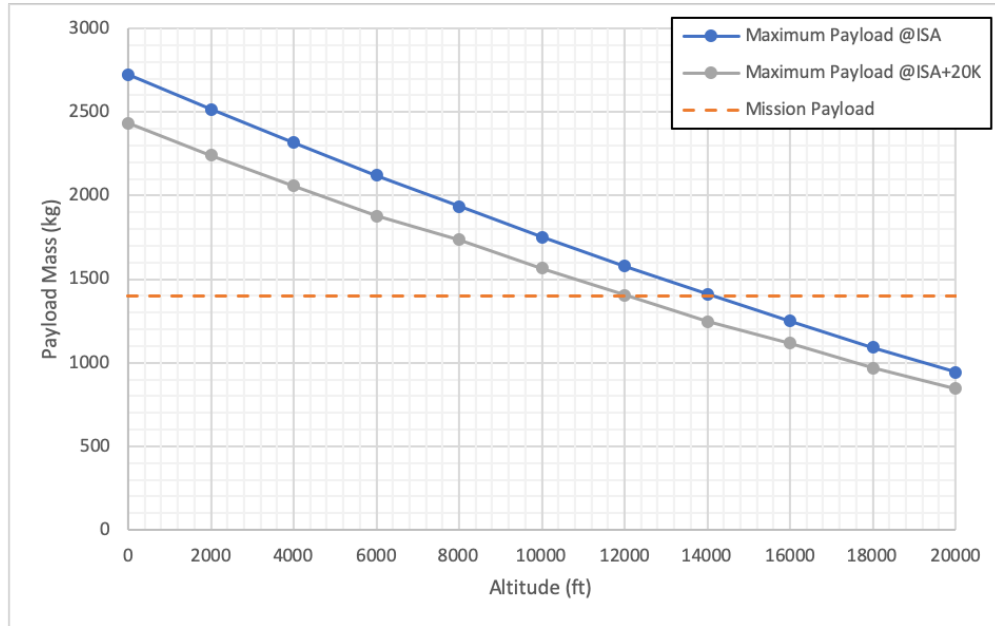


Figure 9: Payload Mass vs Hover Ceiling for our helicopter.

The results show that our helicopter is capable of carrying the mission payload in hover up to an altitude of 14,000 ft during ideal weather conditions (i.e. ISA temperature) while it could carry the mission payload in hover up to a lower altitude when conditions are less ideal (capabilities at ISA temperature + 20K is shown in Figure 9). If the helicopter were to be used for other purposes, it could still carry a payload of around 1 tonne at 20,000 ft.

2 Other Contributions

This section outlines the other notable contributions I have made to the project, which includes assisting my colleagues within the WRR05 team for flight mechanics and control, and consulting for colleagues in other teams that would like opinions from a flight mechanics standpoint.

2.1 Free-Body Diagram

After the team has decided to go with a helicopter with twin-intermeshing-rotors, the Flight Mechanics requires an aerodynamic model for the helicopter to begin the state space formulation. Hence, I formulated a Free-Body Diagram for the team to use for future analyses of the helicopter. The Free-Body Diagram is shown in Appendix A.

2.2 State Space Model Formulation: Jacobian of ODE for Blade Flapping Angle

The blade flapping angle (β) is considered to be one of the states of the helicopter system, and since it is described by a second-order ODE in equation 3.34 of Padfield [15], its Jacobian according to all of the states has to be found for it to be incorporated into the state matrix. I found the Jacobian by using symbolic differentiation in MATLAB. The full results are shown in Appendix B.

2.3 Empennage Sizing

I was invited to consult and verify for the methodology of the sizing the H-tail of our helicopter, and to comment on its effects on the helicopter's stability. The details of this sizing is in the report of my colleague from the aerodynamics team [18].

2.4 LQG vs MPC Control Trade-off study

The WRR05 team has to decide on a method of control for our helicopter control system. Upon consultation with Dr Andrew Wynn. We decided on the choice of either using Linear-Quadratic-Gaussian (LQG) control or Model Predictive Control (MPC). I along with another colleague in the WRR05 team performed a trade-off study between the two types on controllers. I was in charge of the research of LQG control, hence I'll briefly detail the reasons on why I think LQG is a superior choice. Details on MPC is in my colleague's report [19].

Firstly, LQG control is relatively easy to implement, as most of the team members are already familiar with LQR control from AERO96002 Control Systems, and for some team members, from AERO96006 Advanced Flight Mechanics. Since LQG is just adding a Kalman filter to LQR, the conceptual jump between LQG and LQR should not be large. Secondly, LQG controllers are good at disturbance rejection due to the presence of the Kalman filter. Thirdly, there are many examples of LQG control being used for controllers with good robustness and performance properties [20], which fulfills the objective of our design. Fourthly, LQG control design is also present as a function in MATLAB. This ensures ease in the design of the controller and better integration with the rest of the code used in the project. Last but not least, there are plentiful resources on the solutions to LQG control problems and the implementation of LQG control to helicopter control systems. The ease of access to resources is a special priority given the current extraordinary circumstances.

2.5 Fuselage Design

I was invited to consult for the design of the fuselage, and I offered my opinion on the optimal fuselage shape based on the C_M results of the CFD models. The details of this sizing is in the report of my colleague from the aerodynamics team [8].

3 Discussion

3.1 Evaluation of Methodology

Throughout this report, many empirical methods and first order approximations are used. There are many limitations to using such methods including the data only being usable for future reference instead of directly by pilots and operators due to inaccuracies. However, given that we are in the first design iteration and that there are many aspects of the data regarding engine performance, drag and exact flight trajectory that are lacking to perform a complete performance analysis, using first order approximations is the best that we could do for now to create reasonably accurate data for future reference, as opposed to using fully analytical methods to do so.

Needless to say, once a more complete and accurate set of testing and performance data are available, the performance metrics should be updated in terms of completeness and accuracy in accordance as well.

3.2 Recommendations for Future Designs

It is found that the power of the engine sized by the propulsion team is for the helicopter to take off at MTOW (i.e. with the full mission payload) instead of GTOW (i.e. without the payload), resulting in a much too powerful engine being put into our helicopter. This would result in wasted mass and space being devoted to the engine and would be detrimental to the fuel economy. Furthermore, the fuel tanks are also found to have more volume than is required for our mission. While these certainly makes our helicopters more versatile in terms of other mission capabilities, the first priority of our helicopter design should be tailored to our specific mission profile instead of worrying about its versatility and capability of future missions.

The main hindrance to the performance analysis process itself, especially the analysis involving flight power, is the disagreement between the power values given by the aerodynamics team using their rotor BEM code and engine performance given by the propulsion team. It is later found that the main source of discrepancy is their vastly different assumptions regarding the power generated by the rotors in the overlapping regions, with the aerodynamics team assuming there is no power lapse and the propulsion team assuming there is a power penalty. Although the values obtained by both teams are much more similar after adjusting their assumptions to match one another, this has caused considerable amounts of confusion to both teams and I during the initial stages of the analysis.

Furthermore, it is also recommended that a more comprehensive study on static stability and dynamic stability should be carried out in the future, as with the current data set produced, it is not enough information to deduce the limiting speed after which the helicopter-payload system becomes unstable during 2nd cruise due to a lack of time.

4 Conclusion

4.1 Summary of Performance Metrics

Table 4: Summary of Performance Metrics.

Performance Metric	Value	Remarks
Max Rate of Climb at GTOW	37.74 m s ⁻¹	
Best Climb Speed at GTOW	20.12 m s ⁻¹	
Max Rate of Climb at MTOW	16.89 m s ⁻¹	
Best Climb Speed at MTOW	32.31 m s ⁻¹	
Rate of Descend	9.4 m s ⁻¹	Achievable without using autorotation and without onset of VRS.
Max Forward Flight Speed with slung payload	30 m s ⁻¹	
Best Range Speed GTOW @20,000 ft	62.48 m s ⁻¹	
Best Endurance Speed GTOW @20,000 ft	26.21 m s ⁻¹	
Best Range Speed MTOW @3,000 ft	30 m s ⁻¹	Limited by max forward flight speed with payload.
Best Endurance Speed MTOW @3,000 ft	30 m s ⁻¹	Same as above.
Fuel-to-climb	13.25 kg	
Fuel required by mission	202.55 kg	Initial estimation is 273.7 kg.
Maximum range boost	54.3 km	Using all extra fuel in 2nd cruise.
Maximum endurance boost	1810 s	Using all extra fuel in last hover.
Hover ceiling with mission payload @ISA temp.	14,000 ft	
Hover ceiling with mission payload @ISA+20K temp.	12,000 ft	
Maximum payload @20,000 ft @ISA temp.	942 kg	
Maximum payload @10,000 ft @ISA temp.	1753 kg	

4.2 List of Future Design Recommendations

1. Improve communication between aerodynamics and propulsion team regarding assumptions used in calculation of power and unify the correction factors for power used regarding the overlapping regions of the rotor to reduce discrepancies in values.
2. Reduction in power of engine to reduce weight and size of engine and improve fuel economy, and size the power of the engine to take off at GTOW in lieu of MTOW.
3. Reduction of fuel tank sizes according to the new recommended fuel weights to save space for additional on board equipment or to facilitate a more efficient fuselage design.
4. Conduct a more comprehensive static and dynamic stability analysis in the future to determine maximum speed before the system becomes unstable during 2nd cruise.
5. Use of analytical methods and less first-order approximations in future performance analysis when more complete and accurate data is available.

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A Free-Body Diagram of our helicopter

The section provides the free-body diagrams which the WRR05 team has used to formulate the state-space model of the helicopter, and also serves as the basis of the aerodynamic model that the aerodynamics team used in their performance analysis.

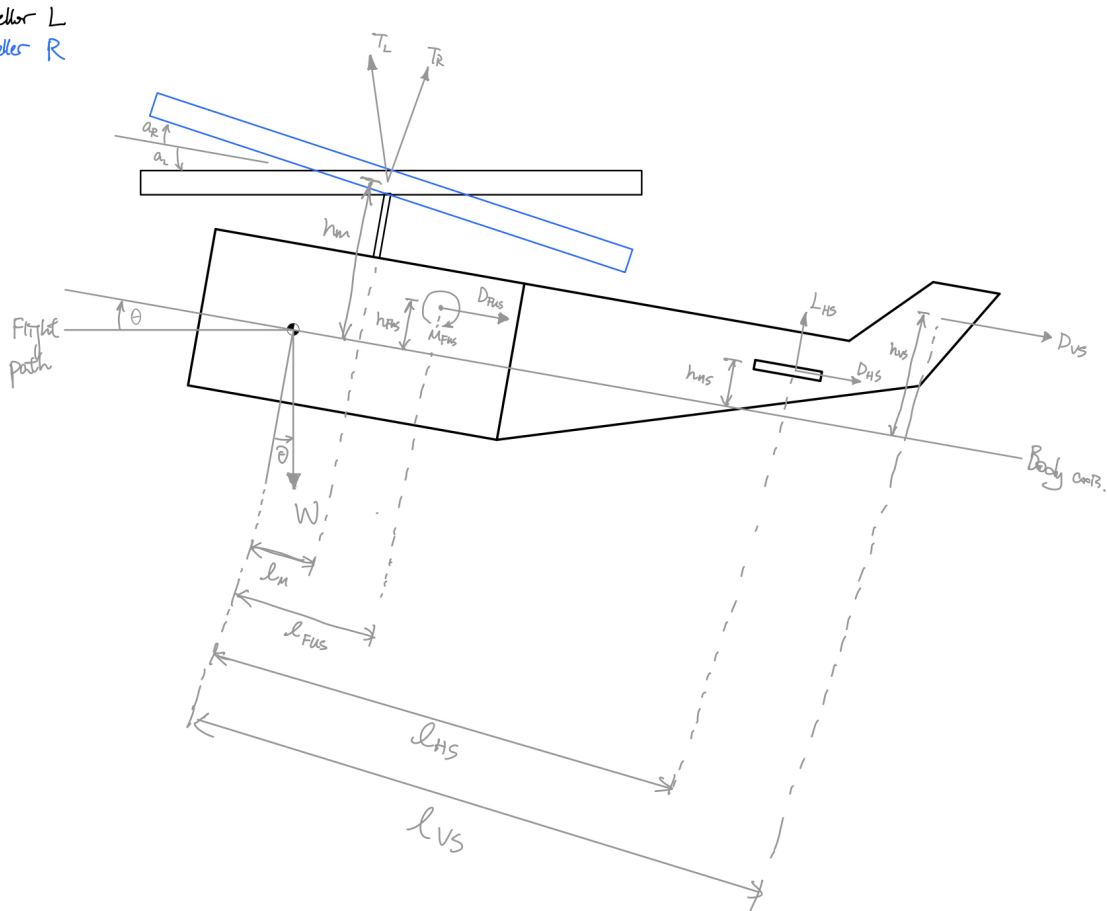


Figure 10: Side view of Free-Body Diagram.

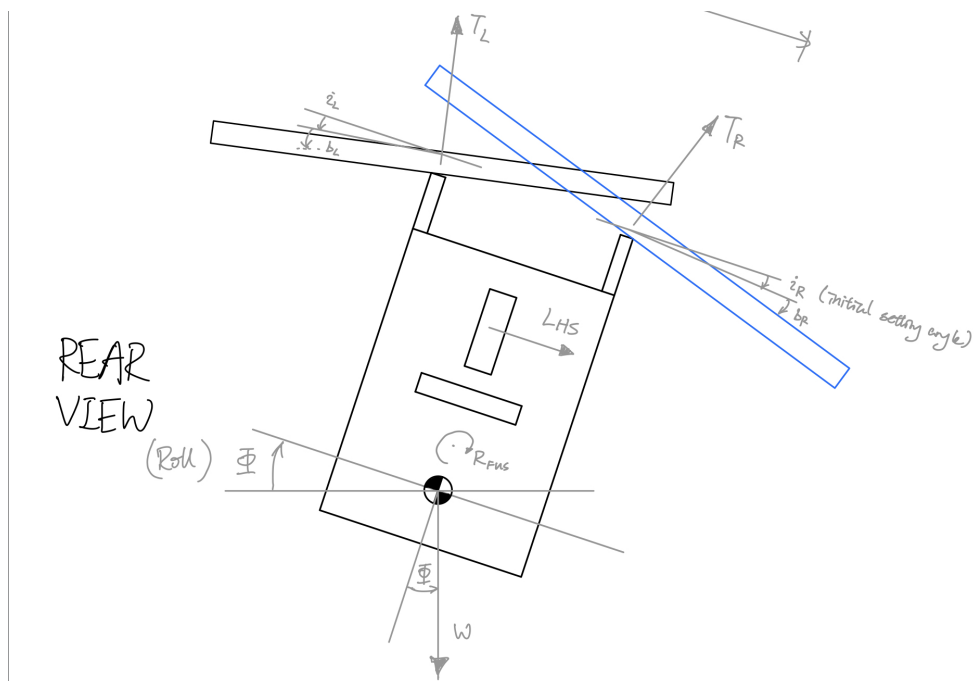


Figure 11: Rear view of Free-Body Diagram.

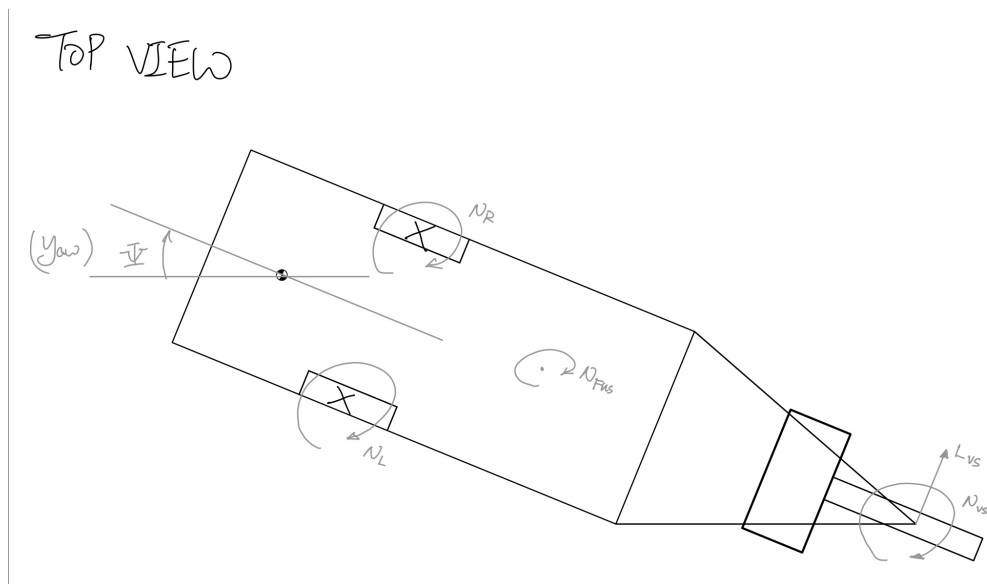


Figure 12: Top view of Free-Body Diagram.

B Full algebraic expression of Jacobian of $\ddot{\beta}$

Both the full results and the MATLAB code used are detailed in this section.

B.1 Results

The general expression for the Jacobian of $\ddot{\beta}$ from equation 3.34 of Padfield is described by

$$J = \left[\ddot{\beta}_u \ddot{\beta}_v \ddot{\beta}_w \ddot{\beta}_p \ddot{\beta}_q \ddot{\beta}_r \ddot{\beta}_\phi \ddot{\beta}_\theta \ddot{\beta}_\psi \ddot{\beta}_{\beta_{1c}} \ddot{\beta}_{\beta_{1c}} \ddot{\beta}_{\beta_{1s}} \ddot{\beta}_{\beta_{1s}} \ddot{\beta}_{\beta_{2c}} \ddot{\beta}_{\beta_{2c}} \ddot{\beta}_{\beta_{2s}} \ddot{\beta}_{\beta_{2s}} \right]^T$$

where

$$\begin{aligned} \ddot{\beta}_u = & \gamma \cdot \left(\theta_P \cdot \left(\frac{u \cdot \sin^2(\psi_i)}{2 \cdot \Omega^2 \cdot R^2} + \frac{u \cdot \sin(\psi_i)}{3 \cdot \Omega^2 \cdot R^2 \cdot \mu} \right) - \lambda_1 \cdot \left(\frac{\sin(\psi_i) \cdot \mu}{6} + \frac{1}{8} \right) - \lambda_0 \cdot \left(\frac{\sin(\psi_i) \cdot \mu}{4} + \frac{1}{6} \right) + \theta_{tw} \cdot \left(\frac{u \cdot \sin^2(\psi_i)}{3 \cdot \Omega^2 \cdot R^2} + \right. \\ & \left. \frac{u \cdot \sin(\psi_i)}{4 \cdot \Omega^2 \cdot R^2 \cdot \mu} + \frac{u \cdot \sin(\psi_i) \cdot (\mu_z - \lambda_0)}{4 \cdot \Omega^2 \cdot R^2 \cdot \mu} + \frac{u \cdot \sin(\psi_i) \cdot (\overline{\omega}_y - \lambda_1)}{6 \cdot \Omega^2 \cdot R^2 \cdot \mu} \right) + 2 \cdot \cos(\psi_i) \cdot \left(\overline{p}_{wu} + \frac{\overline{q}_{w'u}}{2} \right) + 2 \cdot \sin(\psi_i) \cdot \left(-\overline{q}_{wu} + \frac{\overline{p}_{w'}}{2} \right) \\ & - \beta \cdot \frac{\gamma \cdot u}{\Omega^2 \cdot R^2} \cdot \left(\frac{\cos(\psi_i) \cdot \sin(\psi_i)}{4} + \frac{\cos(\psi_i) \cdot \left(\frac{\sin(\psi_i) \cdot \mu}{4} + \frac{1}{6} \right)}{\mu} \right) - \frac{\beta \cdot \gamma \cdot u \cdot \sin(\psi_i)}{6 \cdot \Omega^2 \cdot R^2 \cdot \mu}, \end{aligned}$$

$$\begin{aligned} \ddot{\beta}_v = & \gamma \cdot \left(\theta_P \cdot \left(\frac{v \cdot \sin^2(\psi_i)}{2 \cdot \Omega^2 \cdot R^2} + \frac{v \cdot \sin(\psi_i)}{3 \cdot \Omega^2 \cdot R^2 \cdot \mu} \right) - \lambda_{1v} \cdot \left(\frac{\sin(\psi_i) \cdot \mu}{6} + \frac{1}{8} \right) - \lambda_{0v} \cdot \left(\frac{\sin(\psi_i) \cdot \mu}{4} + \frac{1}{6} \right) + \theta_{tw} \cdot \left(\frac{v \cdot \sin^2(\psi_i)}{3 \cdot \Omega^2 \cdot R^2} + \right. \\ & \left. \frac{v \cdot \sin(\psi_i)}{4 \cdot \Omega^2 \cdot R^2 \cdot \mu} + \frac{v \cdot \sin(\psi_i) \cdot (\mu_z - \lambda_0)}{4 \cdot \Omega^2 \cdot R^2 \cdot \mu} + \frac{v \cdot \sin(\psi_i) \cdot (\overline{\omega}_y - \lambda_1)}{6 \cdot \Omega^2 \cdot R^2 \cdot \mu} \right) + 2 \cdot \cos(\psi_i) \cdot \left(\overline{p}_w + \frac{\overline{q}'_w v_w}{2} \right) + 2 \cdot \sin(\psi_i) \cdot \left(-\overline{q}_w + \frac{\overline{p}_{w'}}{2} \right) \\ & - \beta \cdot \frac{\gamma \cdot v}{\Omega^2 \cdot R^2} \cdot \left(\frac{\cos(\psi_i) \cdot \sin(\psi_i)}{4} + \frac{\cos(\psi_i) \cdot \left(\frac{\sin(\psi_i) \cdot \mu}{4} + \frac{1}{6} \right)}{\mu} \right) - \frac{\beta \cdot \gamma \cdot v \cdot \sin(\psi_i)}{6 \cdot \Omega^2 \cdot R^2 \cdot \mu}, \end{aligned}$$

$$\begin{aligned} \ddot{\beta}_w = & 2 \cdot \cos(\psi_i) \cdot \left(\overline{p}_{ww} + \frac{\overline{q}'_{w'w}}{2} \right) + 2 \cdot \sin(\psi_i) \cdot \left(-\overline{q}_{ww} + \frac{\overline{p}_{w'w}}{2} \right) \\ & - \gamma \cdot \left(\lambda_{1w} \cdot \left(\frac{\sin(\psi_i) \cdot \mu}{6} + \frac{1}{8} \right) + \lambda_{0w} \cdot \left(\frac{\sin(\psi_i) \cdot \mu}{4} + \frac{1}{6} \right) \right), \end{aligned}$$

$$\ddot{\beta}_p = \ddot{\beta}_q = \ddot{\beta}_r = \ddot{\beta}_\phi = \ddot{\beta}_\theta = \ddot{\beta}_\psi = 0.$$

For rotors 1 and 2 in both longitudinal and latitudinal directions, the general expressions of $\ddot{\beta}_\beta$ and $\ddot{\beta}_{\dot{\beta}}$ are the same. Hence, the general expressions are

$$\ddot{\beta}_\beta = -\lambda_\beta^2 - \gamma \cdot \cos(\psi_i) \cdot \mu \cdot \left(\frac{\sin(\psi_i) \cdot \mu}{4} + \frac{1}{6} \right),$$

$$\ddot{\beta}_{\dot{\beta}} = -\gamma \cdot \left(\frac{\sin(\psi_i) \cdot \mu}{6} + \frac{1}{8} \right).$$

Note that $\psi_i = 0$ in the longitudinal direction while $\psi_i = 90$ in the latitudinal direction.

B.2 MATLAB Code

The output of this MATLAB code is detailed in the previous section.

```
1 % Script for GDP
2 clear
3 clc
4 close all
5
6 % syms in question
7 syms beta
8 syms dbeta
9 syms ddbeta
10
11 % other state variables
12 syms u v w p q r phi theta psi
13
14 syms mu(u,v,w)
15 syms mu_z
16 syms gamma
17 syms lambda0(u,v,w)
18 syms lambda1(u,v,w)
19 syms lambda_beta
20 syms omega_y
21 syms qwp(u,v,w)
22 syms qw(u,v,w)
23 syms pwp(u,v,w)
24 syms pw(u,v,w)
25 syms thetap
26 syms thetatw
27
28 syms psi_i
29
30 syms f_betap
31 syms f_beta
32 syms f_lambda
33 syms f_thetap
34 syms f_thetatw
35 syms f_omega
36
37 syms Omega
38 syms rb
39 syms R
40 syms theta
41
42 % define beta differentials
43 %dbeta = diff(beta);
44 %ddbeta = diff(dbeta);
45
46 % define mu
47 mu = sqrt((u^2+v^2)/(Omega*R)^2);
48
49 % define theta
50 % theta = thetap + (rb/R)*thetatw;
51
52 % define all f
53 f_betap = (1/8)*(1 + 4/3*mu*sin(psi_i));
54 f_beta = (1/8)*(4/3 + 2*mu*sin(psi_i));
55 f_lambda = f_beta;
56 f_thetap = (1/8)*(1 + 8/3*mu*sin(psi_i) + 2*mu^2*sin(psi_i)^2);
57 f_thetatw = (1/8)*(4/5 + 2*mu*sin(psi_i) + 4/3*mu^2*sin(psi_i)^2);
```



```

58 f_omega    = f_betap;
59
60 % input flap equation eq(3.34)
61 ddbeta = - f_betap*gamma*dbeta - (lambda_beta^2+gamma*mu*cos(psi_i)*f_beta)*beta + ...
        2*((pw+qwp/2)*cos(psi_i)-(qw-pwp/2)*sin(psi_i)) + ...
        gamma*(f_thetap*thetap+f_thetatw*thetatw+f_lambda*(mu_z-lambda0)+
62 f_omega*(omega_y-lambda1));
63
64 % diff by u
65 A1 = diff(ddbeta,u)
66 A2 = diff(ddbeta,v)
67 A3 = diff(ddbeta,w)
68 A4 = diff(ddbeta,p)
69 A5 = diff(ddbeta,q)
70 A6 = diff(ddbeta,r)
71 A7 = diff(ddbeta,phi)
72 A8 = diff(ddbeta,theta)
73 A9 = diff(ddbeta,psi)
74 A10 = diff(ddbeta,beta)
75 A11 = diff(ddbeta,dbeta)

```

C Autorotation curve in ROD vs Speed diagram

Note that the autorotation curve is the lower boundary of the 'Boundaries by Snapir' in the following figure.

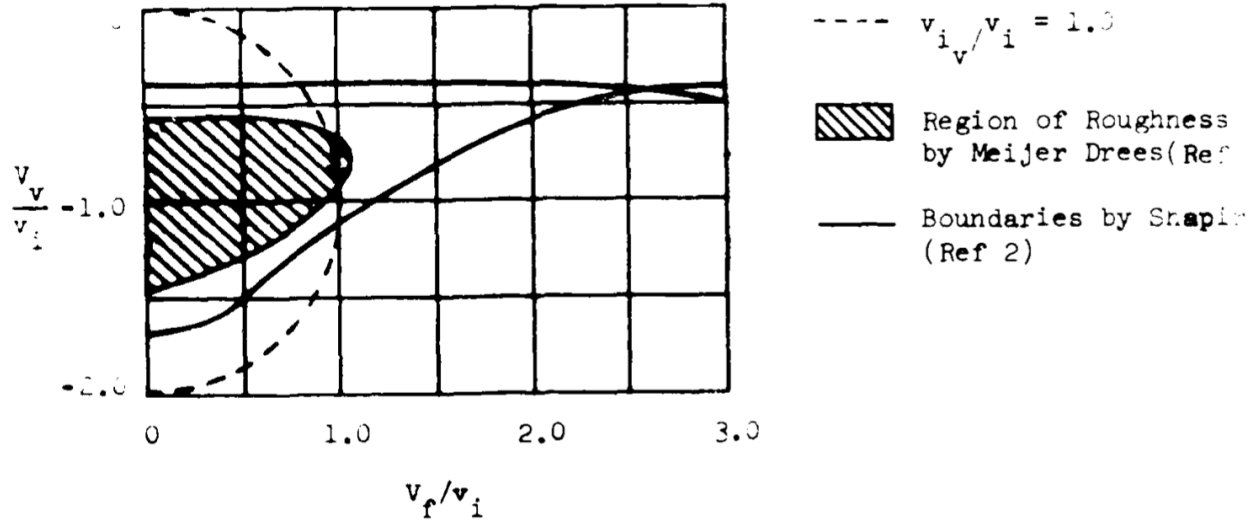


Figure 15
Vortex Ring Region

Figure 13: Autorotation curve in ROD vs Speed diagram [3].

D Full Set of Data from Range and Endurance Analysis

The following table contains the full set of spreadsheet data used in the range and endurance analysis from the initial fuel allocation. Lapse in SFC has already been accounted for and it is found that altitude doesn't have a profound effect on SFC.

Table 5: Range and Endurance Analysis with the fuel allocated (Full spreadsheet data).

Segment	SFC (kg kW ⁻¹ h ⁻¹)	U (m s ⁻¹)	P (kW)	Fuel allocated (kg)	SAR (m/kg)	SE (s/kg)	Range (m)	Endurance (s)
Climb	N/A	N/A	N/A	13.250	N/A	N/A	N/A	N/A
1st Cruise @Best Range Speed	0.298	62.484	373.808		2019.325	32.317	111567.724	1785.541
1st Cruise @Best Endurance Speed	0.298	26.210	251.600	55.250	1258.469	48.015	69530.428	2652.821
Hover for 10 mins	0.298	0.000	410.000	40.200	0.000	29.465	0.000	1184.482
Descend and Capture	0.295	36.230	410.000	56.400	1078.363	29.764	60819.671	1678.710
2nd Cruise	0.298	30.000	474.820	86.500	763.271	25.442	66022.900	2200.763
Hover for 5 mins	0.298	0.000	604.000	22.100	0.000	20.001	0.000	442.020

Below is the same for the new recommended fuel allocation

Table 6: Range and Endurance Analysis with the new fuel allocation (Full spreadsheet data)

Segment	SFC (kg kW ⁻¹ h ⁻¹)	U (m s ⁻¹)	P (kW)	Fuel allocated (kg)	SAR (m/kg)	SE (s/kg)	Range (m)	Endurance (s)
Climb	N/A	N/A	N/A	13.250	N/A	N/A	N/A	N/A
1st Cruise @Best Range Speed	0.298	62.484	373.800	37.150	2019.369	32.318	75019.572	1200.621
Hover for 10 mins	0.298	0.000	410.000	20.380	0.000	29.465	0.000	600.491
Descend and Capture	0.295	36.230	410.000	18.500	1078.363	29.764	19949.715	550.641
2nd Cruise	0.298	30.000	474.820	98.270	763.271	25.442	75006.594	2500.220
Hover for 5 mins	0.298	0.000	604.000	15.000	0.000	20.001	0.000	300.013